

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Nirmal Kasi 192424248 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Nirmal Kasi

TECHNICAL PRESENTATION

1. "Steganography vs Cryptography: Hidden Security in the Digital World"

Steganography vs Cryptography: Hidden Security in the Digital World

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CSA5112 Cryptography And Network Security

Introduction

What is Information Security?

- Information security is the process of protecting digital information from unauthorized access, modification, or theft.
- Two important techniques used are:

Cryptography - Hides the meaning of data by encrypting it.

Steganography - Hides the existence of the data itself.

Objective

- To understand how both techniques work, compare them, and identify their real-world applications.

What is Cryptography?

Definition

- Cryptography is the science of converting readable information (**Plaintext**) into unreadable information (**Ciphertext**) using encryption algorithms.



Goals

- Confidentiality
- Integrity
- Authentication
- Non-repudiation

Examples

- AES
- RSA
- DES
- ECC

Types of Cryptography

Symmetric Key Cryptography

- Same key is used for encryption and decryption.
- Very fast.
- Suitable for large files.

Examples:

- AES
- DES
- Blowfish

Asymmetric Key Cryptography

Uses two keys:

- Public Key
- Private Key

Examples:

- RSA
- ECC

Advantages

- Secure key exchange
- Digital signatures

Types of Cryptography

Hash Functions

- One-way mathematical functions
- Cannot be decrypted
- Used for password storage and integrity verification

Examples

- SHA-256
- SHA-3
- MD5 (obsolete)

What is Steganography

Definition

- Steganography is the technique of hiding secret information inside another file without changing its visible appearance.

The hidden message may be placed inside:

- Images
- Audio files
- Videos
- Text files

Example



Types of Steganography

- **Image Steganography**
- Most commonly used
- Data hidden in image pixels
- Uses Least Significant Bit (LSB)

Audio Steganography

- Secret data hidden inside sound files
- Difficult for humans to notice

Network Steganography

Hides information inside:

- TCP packets
- IP headers
- Network protocols

Audio Steganography

- Secret data hidden inside sound files
- Difficult for humans to notice

Text Steganography

- Uses spaces
- Invisible characters
- Font changes
- Word positioning

Steganography vs Cryptography

Feature	Cryptography	Steganography
Purpose	Hide data meaning	Hide data existence
Visibility	Encrypted data is visible	Hidden data is invisible
Security	High	Moderate
Detection	Easy to detect encrypted file	Difficult to detect
Data Capacity	Unlimited	Limited by cover file
Best Use	Secure communication	Secret communication

Best Practice

Combine both techniques:

1. Encrypt the message.
2. Hide the encrypted message inside an image.

This provides two layers of security.

Applications

Cryptography

- Online Banking
- HTTPS Websites
- Email Security
- VPN
- Digital Signatures
- Cryptocurrency

Steganography

- Military communication
- Copyright protection
- Digital watermarking
- Secret communication
- Medical record security
- Intelligence agencies

Advantages and Limitations

Cryptography

Advantages

- Very secure
- Widely accepted
- Supports authentication
- Protects confidentiality

Limitations

- Encrypted data attracts attention
- Key management is difficult
- Computational overhead

Steganography

Advantages

- Hidden communication
- Difficult to detect
- Maintains normal appearance

Limitations

- Limited storage capacity
- Can be destroyed by image compression
- Easier to break if discovered

Conclusion

- Cryptography protects the **content** of information.
- Steganography protects the **existence** of information.
- Both techniques play an essential role in modern cybersecurity.
- Using cryptography and steganography together provides stronger security than using either one alone.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation